

SQL Interview Question

1. What is SQL and why it is used?

SQL stands for **Structured Query Language**. It's a programming language designed for **managing and manipulating** relational databases. SQL is used to perform various operations on databases such as **storing, retrieving, updating, and deleting data**. It provides a standardized way to interact with databases, making it **easier to manage and analyze large volumes of data** efficiently.

2. Why we need a database?

We need a database to **efficiently store, organize, and retrieve large amounts of structured data**, enabling easier access, manipulation, and analysis of information for various purposes such as **managing business operations, storing user data, and facilitating data-driven decision making**.

3. How does an sql database works?

An SQL database works by organizing data into **tables**, each containing rows and columns. It uses a structured query language (SQL) to interact with the data. Users can perform operations like **inserting, updating, deleting, and querying data** using SQL commands. The database engine manages tasks like **storing and retrieving data efficiently**, ensuring data integrity through constraints, and optimizing queries for performance.

4. What the main usage of SQL?

The main usage of SQL is to **interact with and manage relational databases**. It allows users to perform tasks such as **storing, retrieving, updating, and deleting data** from databases in a structured and efficient manner.

5. What do you understand by tables and fields in a database?

The main usage of SQL is to interact with and manage relational databases. It allows users to perform tasks such as storing, retrieving, updating, and deleting data from databases in a structured and efficient manner.

6. Explain different types of commands in SQL

In SQL, there are mainly three types of commands:

- **Data Manipulation Language (DML) commands:** These commands are used to manipulate data stored in the database. Common DML commands include **SELECT** (retrieve data), **INSERT** (insert new data), **UPDATE** (modify existing data), and **DELETE** (remove data).
- **Data Definition Language (DDL) commands:** These commands are used to define and manage the structure of the database objects. Common DDL commands include **CREATE** (create new objects like tables, indexes, etc.), **ALTER** (modify existing objects), and **DROP** (delete objects).
- **Data Control Language (DCL) commands:** These commands are used to control access to data within the database. Common DCL commands include **GRANT** (give privileges to users), **REVOKE** (take away privileges from users), etc.

7. What are the statements used under DML commands?

Under Data Manipulation Language (DML) commands, you commonly use statements like:

- **SELECT:** **Retrieves** data from a database.
- **INSERT:** **Adds** new records to a table.
- **UPDATE:** **Modifies** existing records in a table.
- **DELETE:** **Removes** records from a table.

8. What are aggregate function in SQL?

Aggregate functions in SQL are used to perform calculations on **groups of rows to return a single result**. Common aggregate functions include **SUM(), AVG(), MAX(), MIN(), and COUNT()**. They help summarize data by performing operations like **adding up values, finding averages, or identifying the highest or lowest values** within a dataset.

9. What are constraints in SQL?

Constraints in SQL are rules or conditions applied to a table column to ensure **data integrity and accuracy**. They enforce limits or restrictions on the **type, format, or range of values** that can be stored in a column. Examples include **NOT NULL** (requiring a column to have a value), **UNIQUE** (ensuring each value in a column is unique), **PRIMARY KEY** (defining a unique identifier for each row), **FOREIGN KEY** (establishing a relationship between tables), and **CHECK** (specifying conditions that must be met for data to be valid).

10. What are keys in SQL and why do we use them and its types?

Keys in SQL are columns or combinations of columns that are used to uniquely identify **each row in a table**. They ensure data integrity and provide a way to establish relationships between tables. There are several types of keys:

- **Primary Key:** A primary key uniquely identifies each row in a table. It must contain **unique values and cannot contain NULL values**.
- **Foreign Key:** A foreign key is a column or a set of columns in one table that refers to the primary key in another table. It establishes a **link between two tables and enforces referential integrity**.
- **Unique Key:** A unique key ensures that all values in a column (or a combination of columns) are unique. **Unlike a primary key, it can contain NULL values**.
- **Candidate Key:** A candidate key is a column or a set of columns that can uniquely **identify a row** in a table. It can be **chosen as the primary key**.

These keys help maintain data consistency, enforce relationships between tables, and optimize database performance.

11. What are some common clauses used with select query in SQL?

Some common clauses used with SELECT queries in SQL are:

- **WHERE:** **Filters rows** based on a specified condition.
- **ORDER BY:** **Sorts the result** set by one or more columns.
- **GROUP BY:** Groups rows that have the **same values into summary rows**.
- **HAVING:** **Filters groups** based on a specified condition.
- **LIMIT:** **Limits the number of rows** returned in the result set.
- **OFFSET:** Specifies the **number of rows to skip** before starting to return rows.
- **DISTINCT:** Returns only **unique values** in the result set.

12. What is DBMS?

DBMS stands for **Database Management System**. It's software that enables users to interact with databases, allowing them to **store, retrieve, update, and manage data efficiently and securely**.

13. Difference between SQL Vs DBMS?

- **SQL (Structured Query Language)** is a language used for managing and querying relational databases. It's used to perform operations like **retrieving, inserting, updating, and deleting data** from a database.
- **DBMS (Database Management System)** is a software system that provides an interface for users to interact with databases. It handles tasks like **data storage, retrieval, and management, ensuring data integrity and security**. SQL is one of the languages used to interact with a DBMS. So, in short, SQL is a language for querying databases, while DBMS is the software that manages databases.

14. What is Subquery in SQL?

A subquery in SQL is a query **nested within another query**. It's used to **retrieve data from one or more tables** based on a condition specified within the subquery. The result of the subquery can then be used as a **condition or value** in the outer query.

15. What is the SELECT Statement Role in SQL?

The SELECT statement in SQL is used to **retrieve data from a database**. It allows you to specify which columns you want to retrieve and which conditions should be met for the rows you want to retrieve. In essence, it's like asking the database a question and **getting back the relevant information**.

16. What are the Subsets of SQL?

- **DDL** (Data Definition Language): Used to define the structure of the database, including **creating, altering, and dropping** database objects like **tables, indexes, and views**.
- **DML** (Data Manipulation Language): Used to manipulate data within the database, including operations like **INSERT, UPDATE, DELETE, and SELECT**.
- **DCL** (Data Control Language): Used to control access to data within the database, including operations like **GRANT and REVOKE**.
- **TCL** (Transaction Control Language): Used to manage transactions within the database, including operations like **COMMIT, ROLLBACK, and SAVEPOINT**.

17. What is a Primary Key in SQL?

A primary key in SQL is a **unique identifier** for each record in a table. It ensures that each row in the table is **uniquely identifiable and serves as a reference point** for relationships between tables.

18. What is JOINT in SQL and Explain any 2 Types?

In SQL, a JOIN is used to **combine rows from two or more tables** based on a related column between them. Two common types of JOINS are:

- **INNER JOIN:** Retrieves **rows from both tables** where the join condition is met, meaning **only rows with matching values** in both tables are returned.
- **LEFT JOIN:** Retrieves **all rows from the left table** (the first specified table) and **the matched rows from the right table** (the second specified table). If there's no match, **NULL values** are returned for the columns from the right table.

19. What is a Foreign Key?

A foreign key is a column or set of columns in a database table that establishes a relationship with **another table**. It enforces referential integrity by ensuring that the values in the **foreign key column(s)** **correspond to values in the primary key column(s)** of another table.

20. What is the full form of SQL?

SQL stands for **Structured Query Language**.

21. What command do we use to create a new table in SQL?

To create a new table in SQL, you use the **CREATE TABLE** command.

22. Differentiate between SQL and MySQL

- SQL stands for Structured Query Language, which is a standard language for **managing relational databases**. It is used to interact with databases to perform tasks like **querying data, updating data, and defining data structures**.
- MySQL, on the other hand, is a specific **relational database management system** (RDBMS) that uses SQL as its querying language. MySQL is one of the most popular open-source RDBMS, known for its **reliability, ease of use, and scalability**. So, in simple terms, SQL is a language for managing databases, while MySQL is a **specific software implementation** of an RDBMS that uses SQL.

23. Explain the basic structure of an SQL query.

An SQL query typically follows this basic structure:

- **SELECT: Determines** which columns of data you want to retrieve.
- **FROM: Specifies the table** from which you want to retrieve the data.
- **WHERE: Filters the rows** based on specified conditions.
- **GROUP BY: Groups the rows** based on specified columns.
- **HAVING: Filters the grouped rows** based on specified conditions.
- **ORDER BY: Sorts the result** set based on specified columns.
- **LIMIT: Specifies the maximum number** of rows to return (optional).

24. What is an index, and why is it used in databases?

An SQL query typically follows this basic structure:

An index in a database is like a **quick-reference guide for data**, making it **faster to retrieve information**. It's used to optimize search queries by **sorting and organizing** data based on specific columns. This helps speed up data retrieval operations, making database searches **more efficient**.

25. Describe the difference between INNER JOIN and LEFT JOIN.

INNER JOIN returns **only the rows** that have matching values in both tables based on the specified condition, while LEFT JOIN returns **all the rows from the left table and the matched rows from the right table**.

26. How do you add a new record to a table?

To add a new record to a table, you typically use an **SQL INSERT statement**.

27. How can you create empty tables with the same structure as another table?

You can create empty tables with the same structure as another table by using the **"CREATE TABLE ... LIKE"** statement in SQL. This statement **copies the structure of the existing table without copying any data**.

28. Explain database white box testing and black box testing?

- Database **white box testing** involves examining the internal structures and code of a database system to ensure its **correctness, efficiency, and security**. This type of testing focuses on the **logic, algorithms, and database** schema to uncover any bugs or issues within the database implementation.
- **Black box testing**, on the other hand, treats the database system as a closed entity, where the tester **interacts with the system through its inputs and observes the outputs**. It doesn't require knowledge of the internal workings of the database. The goal of black box testing is to **validate the functionality and behavior** of the database system according to its specifications, without considering its internal implementation details.

29. What is the difference between the DROP and TRUNCATE commands?

DROP command is used to **completely remove a table** from a database along with all its data and structure, whereas TRUNCATE command is used to **remove all rows from a table**, but it keeps the table structure intact.

30. What are the benefits of SQL database over NoSQL database?

SQL databases offer **structured data storage, ensuring data consistency and integrity** through predefined schemas. They excel in complex queries and transactions, making them ideal for applications requiring **ACID properties**. NoSQL databases, on the other hand, are **flexible, scalable, and suitable for unstructured or semi-structured data**. They handle large volumes of data and are adept at distributed computing. Choosing between them depends on the specific needs of the project, with SQL databases favored for transactional systems and NoSQL databases for **handling big data and scalability**.

31. What is a constraint, and why use constraints?

A constraint is a **limitation or restriction** placed on a system or process. We use constraints to **guide and control** various aspects of a project or process, ensuring it stays within predefined boundaries and achieves specific goals effectively. Constraints help in managing resources **efficiently, improving focus, and delivering results** within set parameters.

32. What is a SQL operator?

A SQL operator is a **symbol or keyword** used to perform operations on data in a database, such as **comparison, arithmetic, or logical operations**. Examples include "=", "<>", "AND", "OR", "+", "-", "*", and "/".

33. What is the purpose of the GROUP BY clause in SQL?

The purpose of the GROUP BY clause in SQL is to **aggregate data based on specific columns**, allowing you to perform operations like **counting, summing, averaging, or finding minimum and maximum values** within groups of data.

34. How can you prevent SQL injection in your queries?

To prevent SQL injection, use parameterized queries or prepared statements, which **separate SQL code from user input**, making it impossible for attackers to **inject malicious code**.

35. What are stored procedures, and how are they different from functions?

Stored procedures are precompiled SQL code that is **stored in a database and can be executed on demand**. They are different from functions in that stored procedures can perform a **series of SQL statements and other procedural logic**, whereas functions typically **return a single value** and are used within SQL statements.

36. Describe the ACID properties in the context of database transactions.

ACID stands for **Atomicity, Consistency, Isolation, and Durability**. It's a set of properties that ensure **reliability and consistency** in database transactions:

- Atomicity ensures that either **all operations** in a transaction are **completed successfully**, or none are.
- Consistency guarantees that the database remains in a valid state **before and after the transaction**.
- Isolation ensures that the operations within a transaction are isolated from **other concurrent transactions** until the transaction is completed.
- Durability ensures that once a **transaction is committed**, its effects are **permanent and survive system failures**.

37. Explain the use of the HAVING clause in SQL.

The HAVING clause in SQL is used with the GROUP BY clause to filter rows returned by a **query based on aggregate functions**. It allows you to apply conditions to groups created by the **GROUP BY clause**.

38. Differentiate between UNION and UNION ALL.

UNION combines and displays **only distinct rows** from the result sets of **two or more SELECT statements**, while UNION ALL combines **all rows** from the result sets, **including duplicates**.

39. What is the purpose of the COALESCE function?

The purpose of the COALESCE function is to **return the first non-null value** from a list of arguments.

40. How do you perform a self-join in SQL?

In SQL, a self-join is performed by referencing the **same table twice in the query and aliasing** each instance to distinguish between them. This allows you to compare rows within the **same table**.

41. What is the purpose of the window functions in SQL?

Window functions in SQL serve the purpose of **performing calculations** across a set of rows related to the **current row within a query result**, without altering the result set.

42. Explain the differences between clustered and non-clustered indexes.

Clustered indexes determine the **physical order of data rows in a table**, while non-clustered indexes create a **separate structure that points to the data rows**.

43. What is the difference between a view and a table?

A table is a database object that stores data in rows and columns, like a **spreadsheet**. A view, on the other hand, is a **virtual table generated by a query**. It **doesn't store data itself** but presents data from **one or more tables** based on the query criteria.

44. Describe the process of database sharding.

Database sharding is the process of horizontally **partitioning data across multiple servers**. Each partition, or shard, contains a **subset** of the data. This helps distribute the **load and improve scalability** of the database system.

45. Explain the use of the APPLY operator in SQL.

The APPLY operator in SQL is used to **join** a table-valued function with other tables in a query. It's handy for cases where you need to **call a function for each row in a table and incorporate** its results into your query.

46. How do you sort records in a table?

You can sort records in a table by using the ORDER BY clause in a SQL query.

For example:

SELECT * FROM table_name ORDER BY column_name;

This will sort the records in ascending order based on the specified column. If you want to sort in descending order, you can use:

SELECT * FROM table_name ORDER BY column_name DESC;

47. What is the DISTINCT statement, and how do you use it?

The DISTINCT statement in SQL is used to **remove duplicate records** from a result set. You simply include the keyword **DISTINCT immediately after the SELECT keyword** in your SQL query to ensure that only unique rows are returned.

48. What are entities? Give some examples.

Entities are objects or concepts that can be **uniquely identified and have attributes or characteristics**. Examples include people (e.g., John Smith), places (e.g., New York City), organizations (e.g., Google), products (e.g., iPhone), and events (e.g., birthday party).

49. How do we prevent duplicate records when making a query?

To prevent duplicate records when making a query, you can use the **DISTINCT keyword in your SQL query**. This ensures that only unique records are returned, **eliminating duplicates**.

50. How do you add a record to a table?

To add a record to a table, you typically use the **SQL INSERT statement** followed by the values you want to add into each column of the table.

51. Write a query to extract the username(characters before @ symbol) from the Email_ID column.

Here's a simple SQL query to extract the username from the **Email_ID column**:

```
SELECT SUBSTRING_INDEX(Email_ID, '@', 1) AS Username  
FROM your_table_name;
```

52. Write a query to extract domain name like .com, .in, .au etc. from the Email_ID column.

Here's a simple SQL query to extract domain names from the **Email_ID column**:

```
SELECT SUBSTRING_INDEX(Email_ID, '@', -1) AS  
Domain_Name FROM your_table_name;
```

This query will extract the domain name from the Email_ID column by splitting the email address at the '@' symbol and selecting the part after it.

53. Write a query to extract email service provider names like google, yahoo, outlook, etc. from the Email_ID column.

Here's a simple SQL query to extract email service provider names from the **Email_ID column**:

```
SELECT DISTINCT  
SUBSTRING_INDEX(SUBSTRING_INDEX(Email_ID, '@', -1), '.', 1)  
AS Email_Service_Provider  
FROM YourTableName
```

This query will extract the email service provider names like Google, Yahoo, Outlook, etc., from the Email_ID column.

54. What is a view?

A view is a saved SQL query stored in a database, representing a **subset of data from one or more tables**. It appears like a table to users, allowing them to query it **without** knowing the underlying **complexity** of the database structure.

55. Define a temp table?

A temporary table is a database table that exists temporarily and is typically used for **storing intermediate or temporary data** within a database session. It is often used to **simplify complex queries or to store results temporarily** during data processing. Temporary tables are **automatically dropped** when the database session ends or when they are explicitly dropped by the user.

56. What Is primary key?

A primary key is a unique identifier for each record in a database table, ensuring that **no two records have the same value** for the primary key column. It's essential for **data integrity and efficient retrieval** of information.

57. What is default?

The default is a **pre-set value or option** automatically used when **no alternative** is specified.

58. What is foreign key?

A foreign key is a column or a set of columns in a database table that establishes a **link or relationship with a primary key** in another table. It ensures **referential integrity** and helps maintain **consistency** and relationships between tables in a relational database.

59. What is normalisation?

Normalization is the process of organizing data in a database efficiently to **reduce redundancy and dependency, ensuring data integrity and minimizing anomalies.**

60. What is difference between truncate and delete?

Truncate removes **all rows from a table, resetting identity columns**, while delete removes specific rows based on a condition, preserving table **structure and identities.**

61. What is the difference between data definition language (DDL) and data manipulation language (DML)?

Data Definition Language (DDL) is used to define the structure of a database, such as **creating, altering, and dropping tables, indexes, and views.** Data Manipulation Language (DML) is used to manipulate the data within the database, such as **inserting, updating, deleting, and querying data.**

62. What is difference between join and union?

In SQL, JOIN **combines rows from two or more tables** based on a related column, while UNION **combines the results of two or more SELECT statements** into a single result set, removing duplicate rows.

63. Discuss inner join on where clause (with multiple FROM tables)

An inner join **combines rows from two or more tables** based on a **condition specified** in the WHERE clause, showing only the rows that have **matching values**.

64. Describe the difference between truncate and delete?

Truncate **removes all rows from a table** instantly and resets the identity counter, while **delete removes specific rows** based on conditions and can be rolled back using a transaction

65. What is the difference between UNION and UNION ALL?

UNION **combines and removes duplicate rows** from the result set, while UNION ALL **combines all rows, including duplicates, from the result set**.

66. What is the difference between INNER JOIN, OUTER JOIN, FULL OUTER JOIN?

- **INNER JOIN:** Returns **only the rows** that have **matching values in both tables**.
- **OUTER JOIN:** Retrieves **all records** when there is a **match in either left or right table**.
- **FULL OUTER JOIN:** Retrieves all records when there is a match in either left or right table, including **unmatched rows from both tables**.

67. Find duplicate values in a SQL table.

What are the difference between clustered and a non clustered index?

- To find duplicate values in a SQL table, you can use a **GROUP BY clause** along with the **COUNT() function** to identify rows with a count **greater than one** for the columns containing potential **duplicates**.
- Clustered indexes dictate the **physical order of data** in a table, while non-clustered indexes create a **separate structure** that points back to the original data. In simple terms, clustered indexes **sort and store the data rows** in the table based on their key values, while non-clustered indexes create a **separate structure that holds pointers** to the data rows in the table.

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